8.3 Deep Visual-Semantic Alignments

- this work aims to take a step towards the goal of generating dense descriptions of images (Figure 6)
- the primary challenge towards this goal is in the design of a model that is rich enough to simultaneously reason about contents of images and their representation in the domain of natural language
- additionally, the model should be free of assumptions about specific hard-coded templates, rules, or categories and instead rely on learning from the training data
- the second, practical challenge is that datasets of image captions are available in large quantities, but these descriptions multiplex mentions of several entities whose locations in the images are unknown
- the core insight is that we can leverage these large image-sentence datasets by treating the sentences as weak labels, in which contiguous segments of words correspond to some particular, but unknown location in the image
- the approach is to infer these alignments and use them to learn a generative model of descriptions

8.3 Deep Visual-Semantic Alignments

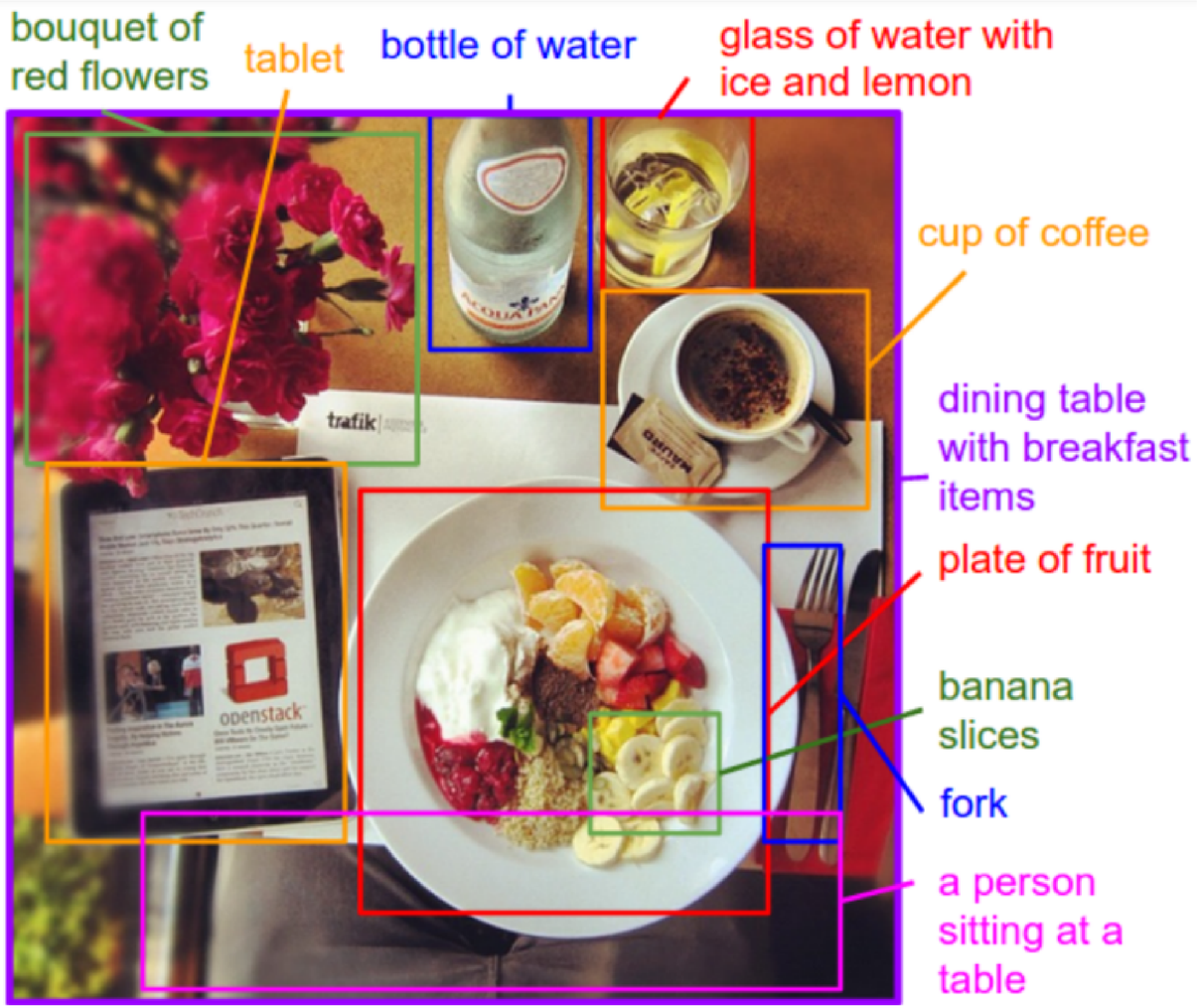


Figure 6: Motivation figure: the model treats language as a rich label space and generates descriptions of image regions.

8.3 Deep Visual-Semantic Alignments

- the ultimate goal of the model is to generate descriptions of image regions
- during training, the input to the model is a set of images and their corresponding sentence descriptions (Figure 7)
- the approach first introduces a model that aligns sentence snippets to the visual regions that they describe through a multimodal embedding
- it then treats these correspondences as training data for a second, multimodal RNN model that learns to generate the snippets
- the alignment model assumes an input dataset of images and their sentence descriptions
- the key insight is that sentences written by people make frequent references to some particular, but unknown location in the image
- for example, in Figure 7, the words “Tabby cat is leaning” refer to the cat, the words “wooden table” refer to the table, etc.
- we would like to infer these latent correspondences, with the eventual goal of later learning to generate these snippets from image regions
- first, neural networks that map words and image regions into a common, multimodal embedding, are introduced
- then a novel objective is introduced, which learns the embedding representations so that semantically similar concepts across the two modalities occupy nearby regions of the space

8.3 Deep Visual-Semantic Alignments

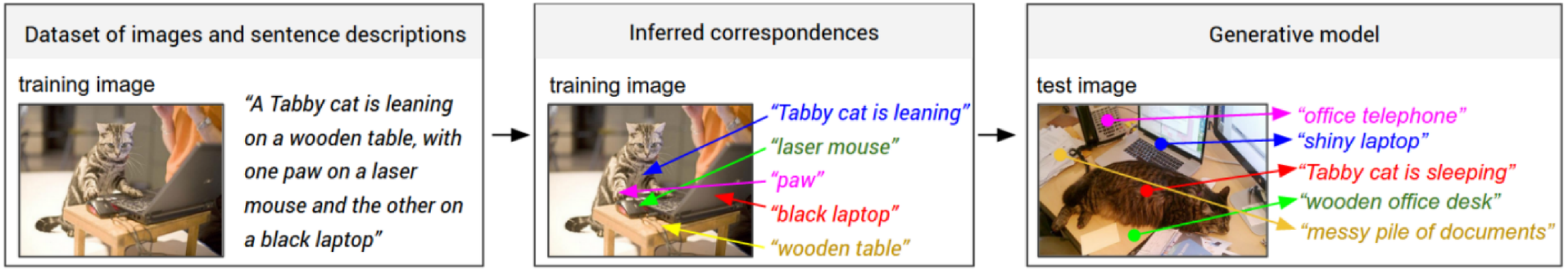


Figure 7: Overview of the approach. A dataset of images and their sentence descriptions is the input to the model (left). The model first infers the correspondences (middle) and then learns to generate novel descriptions (right).

8.3 Deep Visual-Semantic Alignments

- it can be observed that sentence descriptions make frequent references to objects and their attributes
- thus, objects in every image with a Region Convolutional Neural Network (RCNN) are detected
- the CNN is pre-trained on ImageNet
- the top 19 detected locations are used in addition to the whole image and the representations based on the pixels I_b inside each bounding box are computed as follows:

$$v = W_m[CNN_{\theta_c}(I_b)] + b_m,$$

  where $CNN(I_b)$ transforms the pixels inside bounding box I_b into 4096-dimensional activations of the fully-connected layer immediately before the classifier
- the CNN parameters θ_c contain approximately 60 million parameters
- the matrix W_m has dimensions $h \times 4096$, where h is the size of the multimodal embedding space (h ranges from 1000-1600)
- every image is thus represented as a set of h-dimensional vectors $\{v_i | i = 1 \ldots 20\}$

8.3 Deep Visual-Semantic Alignments

- to establish the inter-modal relationships, we would like to represent the words in the sentence in the same *h*-dimensional embedding space that the image regions occupy
- the simplest approach might be to project every individual word directly into this embedding
- however, this approach does not consider any ordering and word context information in the sentence
- to address these concerns, a Bidirectional Recurrent Neural Network (BRNN) is used to compute the word representations
- the BRNN takes a sequence of *N* words (encoded in a 1-of-k representation) and transforms each one into an *h*-dimensional vector
- however, the representation of each word is enriched by a variably-sized context around that word

8.3 Deep Visual-Semantic Alignments

- using the index $t = 1 \ldots N$ to denote the position of a word in a sentence, the precise form of the BRNN is as follows:

$$\begin{aligned} x_t &= W_w \mathbb{I}_t \\ e_t &= f(W_e x_t + b_e) \\ h_t^f &= f(e_t + W_f h_{t-1}^f + b_f) \\ h_t^b &= f(e_t + W_b h_{t+1}^b + b_b) \\ s_t &= f(W_d(h_t^f + h_t^b) + b_d). \end{aligned}$$

- here, $\mathbb{I}_t$ is an indicator column vector that has a single one at the index of the t-th word in a word vocabulary
- the weights W_w specify a 300-dimensional word embedding matrix
- note that the BRNN consists of two independent streams of processing, one moving left to right (h_t^f) and the other right to left (h_t^b) (see Figure 8 for diagram)
- the final h-dimensional representation s_t for the t-th word is a function of both the word at that location and also its surrounding context in the sentence

8.3 Deep Visual-Semantic Alignments

- technically, every s_t is a function of all words in the entire sentence, but the final word representations (s_t) align most strongly to the visual concept of the word at that location ($\mathbb{I}_t$)
- the parameters W_e, W_f, W_b, W_d and the respective biases b_e, b_f, b_b, b_d are learned
- a typical size of the hidden representation ranges between 300-600 dimensions
- the activation function f is ReLU
- we have described the transformations that map every image and sentence into a set of vectors in a common h-dimensional space
- since the supervision is at the level of entire images and sentences, the strategy is to formulate an image-sentence score as a function of the individual region-word scores
- intuitively, a sentence-image pair should have a high matching score if its words have a confident support in the image

8.3 Deep Visual-Semantic Alignments

- the dot product $v_i^\top s_t$ between the i-th region and t-th word can be interpreted as a measure of similarity and we can use it to define the score between image k and sentence l as:

$$S_{kl} = \sum_{t \in g_l} \sum_{i \in g_k} \max(0, v_i^\top s_t).$$

- here, g_k is the set of image fragments in image k and g_l is the set of sentence fragments in sentence l
- the indices k, l range over the images and sentences in the training set
- this score carries the interpretation that a sentence fragment aligns to a subset of the image regions whenever the dot product is positive
- the following reformulation simplifies the model and alleviates the need for additional objectives and their hyperparameters:

$$S_{kl} = \sum_{t \in g_l} \max_{i \in g_k} v_i^\top s_t.$$

- here, every word s_t aligns to the single best image region

8.3 Deep Visual-Semantic Alignments

- assuming that $k = l$ denotes a corresponding image and sentence pair, the final max-margin, structured loss remains:

$$\mathcal{C}(\theta) = \sum_k \left[\underbrace{\sum_l \max(0, S_{kl} - S_{kk} + 1)}_{\text{rank images}} + \underbrace{\sum_l \max(0, S_{lk} - S_{kk} + 1)}_{\text{rank sentences}} \right].$$

- this objective encourages aligned image-sentences pairs to have a higher score than misaligned pairs, by a margin
- now, consider an image from the training set and its corresponding sentence
- we can interpret the quantity $v_i^\top s_t$ as the unnormalized log probability of the t-th word describing any of the bounding boxes in the image
- however, since we are ultimately interested in generating snippets of text instead of single words, we would like to align extended, contiguous sequences of words to a single bounding box

8.3 Deep Visual-Semantic Alignments

- note that the naive solution that assigns each word independently to the highest-scoring region is insufficient because it leads to words getting scattered inconsistently to different regions
- to address this issue, the true alignments are treated as latent variables in a Markov Random Field (MRF), where the binary interactions between neighboring words encourage an alignment to the same region
- concretely, given a sentence with N words and an image with M bounding boxes, we introduce the latent alignment variables $a_j \in \{1 \dots M\}$ for $j = 1 \dots N$ and formulate an MRF in a chain structure along the sentence as follows:

$$E(\boldsymbol{a}) = \sum_{j=1 \dots N} \psi_j^U(a_j) + \sum_{j=1 \dots N-1} \psi_j^B(a_j, a_{j+1}),$$

$$\psi_j^U(a_j = t) = v_i^\top s_t,$$

$$\psi_j^B(a_j, a_{j+1}) = \beta \mathbf{1}_{a_j = a_{j+1}}.$$

8.3 Deep Visual-Semantic Alignments

- here, β is a hyperparameter that controls the affinity towards longer word phrases
- this parameter allows us to interpolate between single-word alignments ($\beta = 0$) and aligning the entire sentence to a single, maximally scoring region when β is large
- the energy is minimized to find the best alignments $\boldsymbol{a}$ using dynamic programming
- the output of this process is a set of image regions annotated with segments of text
- we now describe an approach for generating novel phrases based on these correspondences

8.3 Deep Visual-Semantic Alignments

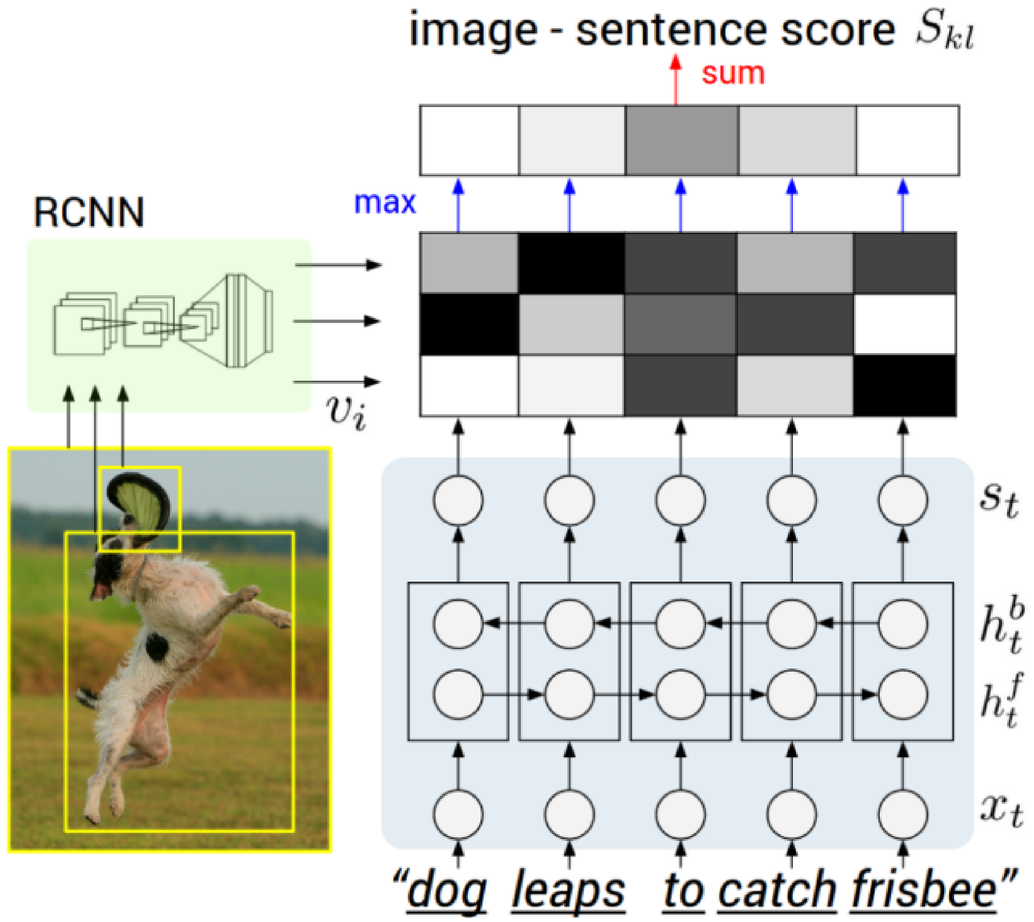


Figure 8: Diagram for evaluating the image-sentence score S_{kl}. Object regions are embedded with a CNN (left). Words (enriched by their context) are embedded in the same multimodal space with a BRNN (right). Pairwise similarities are computed with inner products (magnitudes shown in grayscale) and finally reduced to image-sentence score.

8.3 Deep Visual-Semantic Alignments

- now, we assume an input set of images and their textual descriptions
- these could be full images and their sentence descriptions, or regions and text snippets, as inferred previously
- the key challenge is in the design of a model that can predict a variable-sized sequence of outputs given an image
- in language models based on RNNs, this is achieved by defining a probability distribution of the next word in a sequence given the current word and context from previous time steps
- a simple but effective extension that additionally conditions the generative process on the content of an input image is introduced
- more formally, during training, the multimodal RNN takes the image pixels I and a sequence of input vectors $(x_1, \ldots, x_T)$

8.3 Deep Visual-Semantic Alignments

- it then computes a sequence of hidden states $(h_1, \ldots, h_t)$ and a sequence of outputs $(y_1, \ldots, y_t)$ by iterating the following recurrence relation for $t = 1$ to T:

$$\begin{aligned} b_v &= W_{hi}[CNN_{\theta_c}(I)], \\ h_t &= f(W_{hx}x_t + W_{hh}h_{t-1} + b_h + \mathbf{1}_{t=1} \odot b_v), \\ y_t &= \text{softmax}(W_{oh}h_t + b_o). \end{aligned}$$

- in the equations above, W_{hi}, W_{hx}, W_{hh}, W_{oh}, and b_h, b_o are learnable parameters, and $CNN_{\theta_c}(I)$ is the last layer of a CNN
- the output vector y_t holds the (unnormalized) log probabilities of words in the dictionary and one additional dimension for a special END token
- note that we provide the image context vector b_v to the RNN only at the first iteration
- a typical size of the hidden layer of the RNN is 512 neurons
- the RNN is trained to combine a word (x_t), the previous context (h_{t-1}) to predict the next word (y_t)
- we condition the RNN's predictions on the image information (b_v) via bias interactions on the first step

8.3 Deep Visual-Semantic Alignments

- the training proceeds as follows (see Figure 9): we set $h_0 = \mathbf{0}$, x_1 to a special START vector, and the desired label y_1 as the first word in the sequence
- analogously, we set x_2 to the word vector of the first word and expect the network to predict the second word, etc.
- finally, on the last step, when x_T represents the last word, the target label is set to a special END token
- the cost function is to maximize the log probability assigned to the target labels (i.e., softmax classifier)
- to predict a sentence, we compute the image representation b_v, set $h_0 = \mathbf{0}$, x_1 to the START vector and compute the distribution over the first word y_1
- we sample a word from the distribution (or pick the argmax), set its embedding vector as x_2, and repeat this process until the END token is generated

8.3 Deep Visual-Semantic Alignments

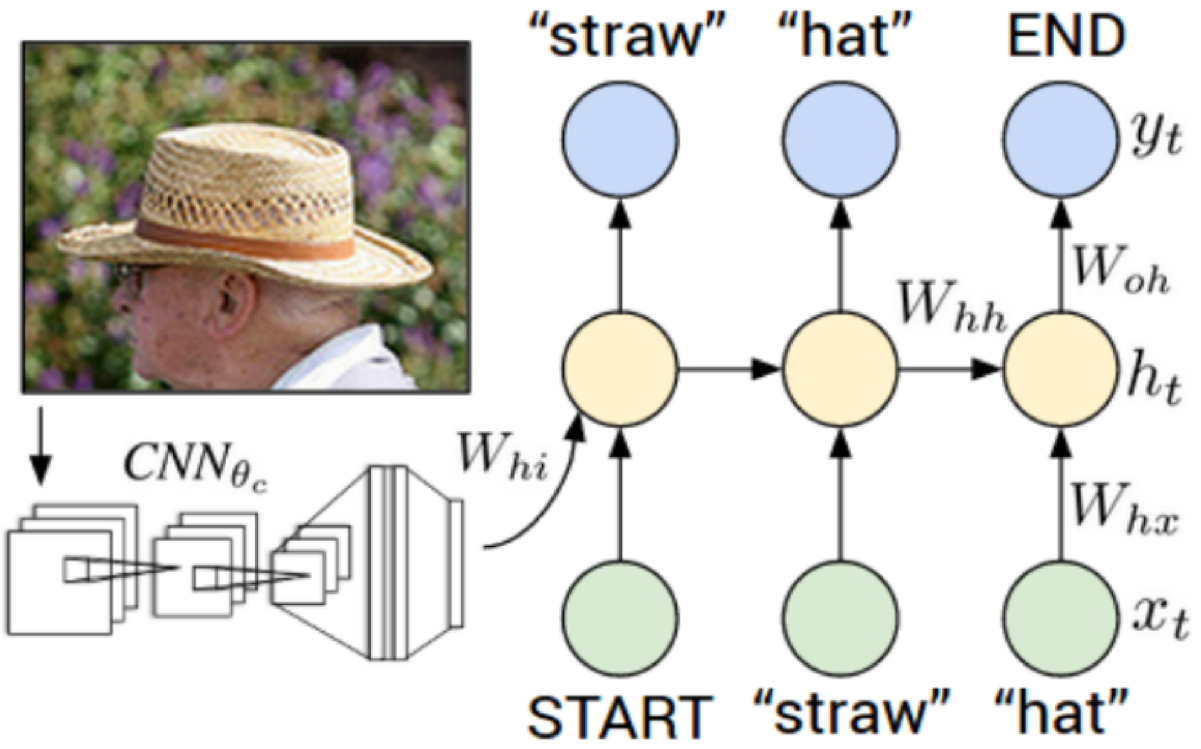


Figure 9: Diagram of the multimodal RNN generative model. The RNN takes a word, the context from previous time steps and defines a distribution over the next word in the sentence. The RNN is conditioned on the image information at the first time step. START and END are special tokens.